03_filtered_indexes

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

PURPOSE

Demonstrate filtered indexes - NC indexes that only include rows matching a WHERE predicate. Useful when a minority value is queried frequently but represents a small fraction of the table. Compares size and seek performance against a full NC index on the same column.

3.0 ENVIRONMENT CHECK

```
USE STN_Lab;
GO

-- ============================================================
==
-- 3.0 -- ENVIRONMENT CHECK
-- Review Estado distribution before creating any index.
-- A filtered index only makes sense for low-frequency value
s.
-- ============================================================
==

SELECT
    Estado,
    COUNT(*)                                          AS total,
    CAST(COUNT(*) * 100.0
         / SUM(COUNT(*)) OVER () AS DECIMAL(5,2))  AS porcent
aje
```

```
FROM dbo.Contribuyente
GROUP BY Estado
ORDER BY total DESC;
GO
```

	Estado	total	porcentaje
1	A	188000	94.00
2	I	10000	5.00
3	S	2000	1.00

3.1 FULL NonClustered INDEX ON Estado

```
--crear indice
CREATE NONCLUSTERED INDEX IX_Contribuyente_Estado_Full
    ON dbo.Contribuyente (Estado)
    INCLUDE (NIT, RazonSocial);
GO

SELECT
    i.name,
    s.used_page_count * 8   AS size_kb,
    s.row_count
FROM sys.dm_db_partition_stats s
JOIN sys.indexes i
    ON s.object_id = i.object_id
   AND s.index_id  = i.index_id
WHERE s.object_id = OBJECT_ID('dbo.Contribuyente')
  AND i.name = 'IX_Contribuyente_Estado_Full';
GO
```

3.2 FILTERED INDEX ON MINORITY Estado VALUE

```
USE STN_Lab;
GO
--crear el indice
CREATE NONCLUSTERED INDEX IX_Contribuyente_Estado_Filtrado
    ON dbo.Contribuyente (Estado)
    INCLUDE (NIT, RazonSocial)
    WHERE Estado = 'S';    -- replace with lowest-frequency v
alue
GO

SELECT
    i.name,
    s.used_page_count * 8   AS size_kb,
    s.row_count
FROM sys.dm_db_partition_stats s
JOIN sys.indexes i
    ON s.object_id = i.object_id
   AND s.index_id  = i.index_id
WHERE s.object_id = OBJECT_ID('dbo.Contribuyente')
  AND i.name = 'IX_Contribuyente_Estado_Filtrado';
GO
```

	name	size_kb	row_count
1	IX_Contribuyente_Estado_Filtrado	104	2000

Query executed successfully. c13\lab_db (16.0 RTM) carlos_roberto (51) STN_Lab 00:00:00 Row: 1, Col: 1 1 rows

OBSERVACION 3.2

- size_kb_filtrado = 104 (debe ser mucho menor que 3.1)
- row_count = 2000 (solo filas donde Estado = 'S')
- Reduccion = 98.8% vs indice completo

3.3 QUERY PERFORMANCE: FULL vs FILTERED INDEX

3.3.1 Forzar indice completo

```
SET STATISTICS IO ON;

SELECT ContribuyenteID, NIT, RazonSocial, Estado
FROM dbo.Contribuyente WITH (INDEX = IX_Contribuyente_Estado_
Full)
WHERE Estado = 'S';

SET STATISTICS IO OFF;
GO

--reads_3.3.1 = 15
```

3.3.2 Forzar indice filtrado

```
SET STATISTICS IO ON;

SELECT ContribuyenteID, NIT, RazonSocial, Estado
FROM dbo.Contribuyente WITH (INDEX = IX_Contribuyente_Estado_
```

```
Filtrado)
WHERE Estado = 'S';

SET STATISTICS IO OFF;
GO

--reads_3.3.2 = 13
--Diferencia vs 3.3.1 = 2
```

3.4 FILTERED INDEX LIMITATION

3.4.1 Funciona predicado coincide con el filtro

Un indice filtrado SOLO funciona cuando el predicado del query coincide exactamente con el filtro del indice. Si el predicado difiere, el optimizador no puede usarlo.

```
SET STATISTICS IO ON;

SELECT ContribuyenteID, NIT, RazonSocial, Estado
FROM dbo.Contribuyente
WHERE Estado = 'S';

SET STATISTICS IO OFF;
GO
```

3.5 OPTIMIZER CHOICE WITHOUT HINTS

```
SET STATISTICS IO ON;

SELECT ContribuyenteID, NIT, RazonSocial, Estado
FROM dbo.Contribuyente
WHERE Estado = 'S';

SET STATISTICS IO OFF;
GO
```

3.7 CLEANUP

```
DROP INDEX IF EXISTS IX_Contribuyente_Estado_Full     ON dbo.
Contribuyente;
DROP INDEX IF EXISTS IX_Contribuyente_Estado_Filtrado ON dbo.
Contribuyente;
```